Problem Set 4: Derivatives

Exercise 1. Dynamic Payoff Replication

1 Data Collection

We consider a European call option on the XSP index with trade date $t_0 = 2026\text{-}04\text{-}17$. The option characteristics are summarized in Table 1.

Variable	Value
S_0 (Underlying price)	712.58
K (Strike)	710
Expiration T	2026-07-31
Bid	28.72
Ask	29.14
$V_0 = \frac{\text{Bid + Ask}}{2}$	28.93

Table 1: XSP option data at trade date t_0.

The risk-free rate is obtained from the 3-month Treasury rate (DGS3MO). Since the value for t_0 is not yet available, we use the latest observation on 2026-04-16:

$$r = 0.0370.$$

Daily SPY prices are used as a proxy for the underlying asset. A sample of approximately one trading year (252 observations) is collected for parameter estimation, and the most recent 30 trading days before t_0 are used as a historical test path.

2 Estimation of GBM Parameters

Log-returns are computed as:

$$R_i = \ln\left(\frac{S_i}{S_{i-1}}\right).$$

Using the annualization factor of 252 trading days, we estimate:

$$\hat{\sigma} = \sqrt{252} \cdot \mathrm{sd}(R_i), \quad \hat{\mu} = 252 \cdot \bar{R} + \frac{1}{2}\hat{\sigma}^2.$$

The resulting estimates are:

$$\hat{\sigma} = 0.1312, \quad \hat{\mu} = 0.3219.$$

Figure 1 shows the 30-day SPY path.

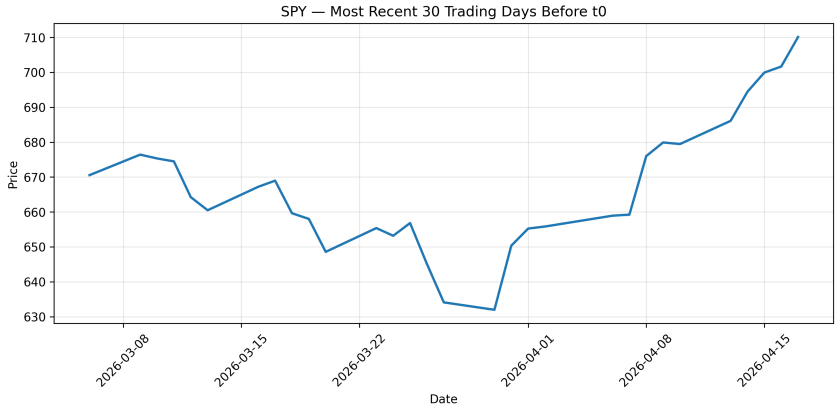


Figure 1: SPY price over the 30 most recent trading days before t_0.

The price path appears continuous and stochastic, consistent with the Geometric Brownian Motion assumption.

Remark on estimation. The drift $\hat{\mu}$ is harder to estimate than volatility $\hat{\sigma}$ because daily returns are dominated by noise, whereas volatility is inferred from dispersion, which is more stable.

3 Implied Volatility and Initial Hedge

The implied volatility $\hat{\sigma}_{imp,0}$ is obtained by solving:

$$C^{BS}(S_0, K, \tau_0, r, \hat{\sigma}_{imp,0}) = V_0.$$

The solution yields:

$$\hat{\sigma}_{imp,0} = 0.1555.$$

The initial hedge parameters are:

$$\Delta_0 = 0.5843, \quad M_0 = V_0 - \Delta_0 S_0 = -387.40.$$

Table 2 summarizes all key parameters.

Parameter	Value
r	0.0370
$\hat{\sigma}$	0.1312
$\hat{\mu}$	0.3219
$\hat{\sigma}_{imp,0}$	0.1555
Δ_0	0.5843
M_0	-387.40

Table 2: Estimated parameters and initial hedge quantities.

4 Self-Financing Hedge and Tracking Error

Using the 30-day historical path, we compute daily Black–Scholes prices $\hat{V}_t$ and deltas Δ_t while keeping K, r, and $\hat{\sigma}_{imp,0}$ fixed.

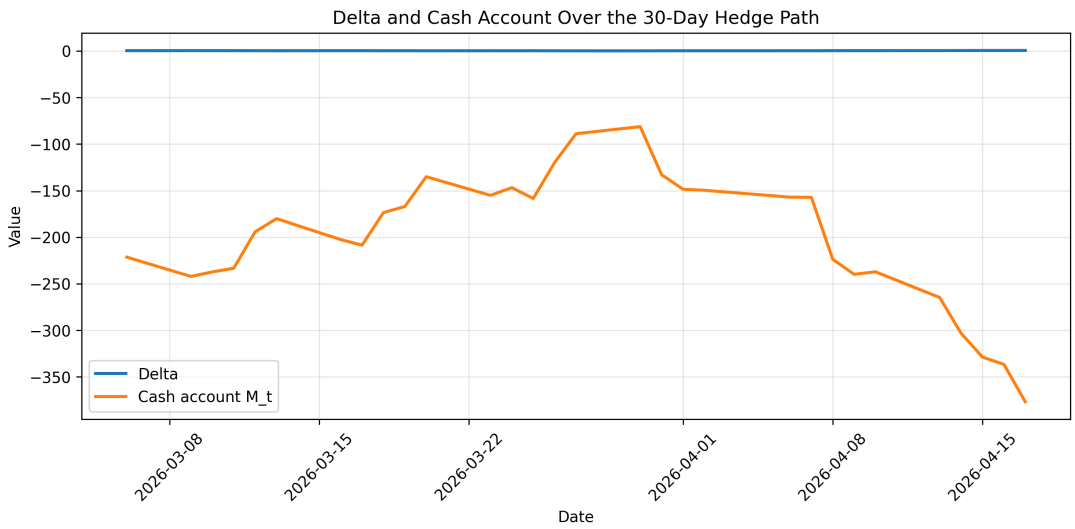
The cash account evolves according to:

$$M_{t_{i+1}} = e^{r\Delta t} M_{t_i} - (\Delta_{t_{i+1}} - \Delta_{t_i}) S_{t_{i+1}}.$$

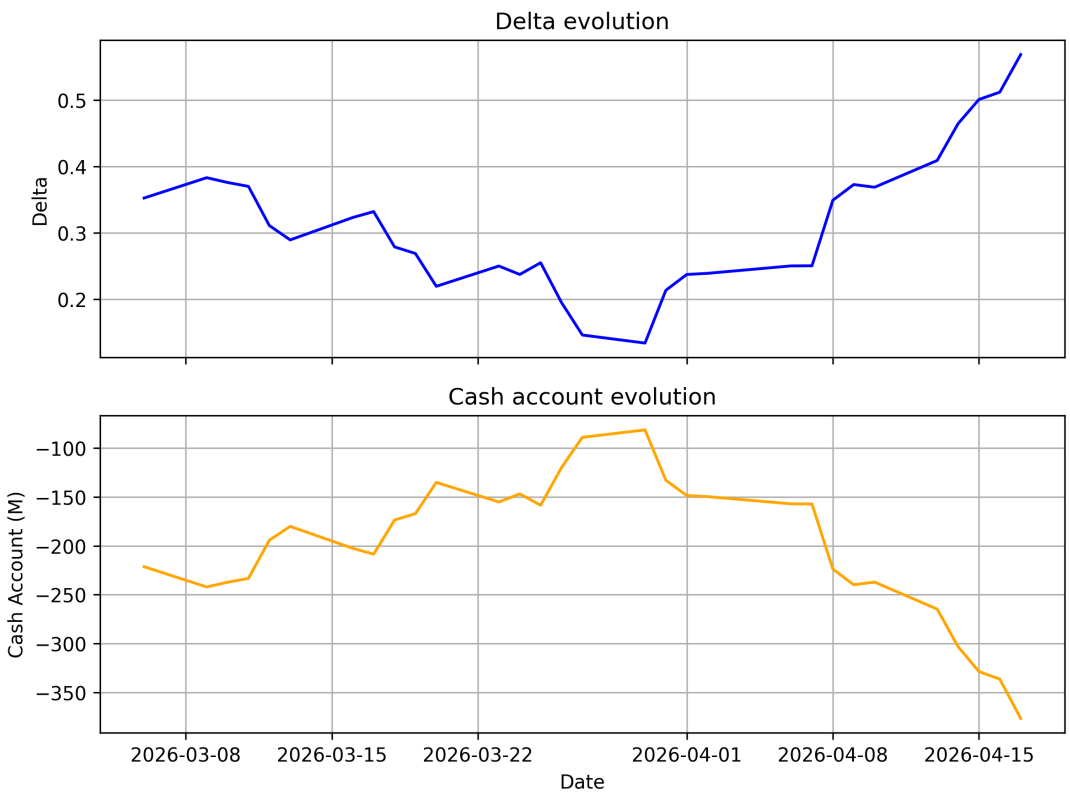
The replicating portfolio is:

$$X_t = \Delta_t S_t + M_t.$$

Figure 2 shows the evolution of Δ_t and M_t.



(a) Combined view of Δ_t and M_t.



(b) Rescaled view showing Δ_t and M_t more clearly.

Figure 2: Evolution of delta and cash account.

Figure 3 compares the replicating portfolio to the theoretical option price.

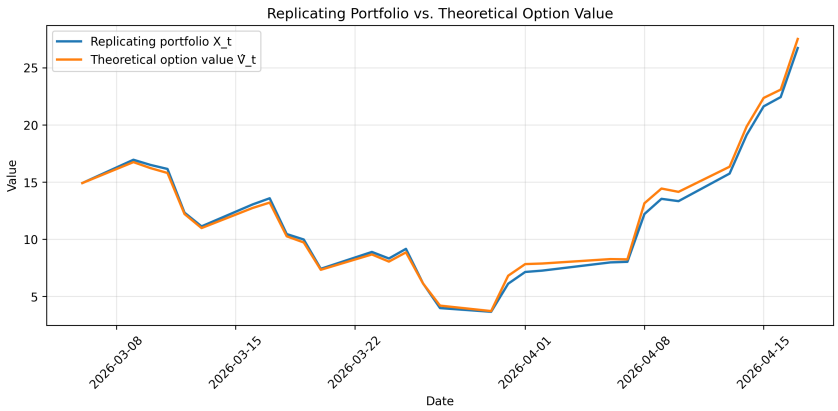


Figure 3: Replicating portfolio X_t vs theoretical price $\hat{V}_t$.

The hedge error is defined as:

$$\text{Error}_t = X_t - \hat{V}_t.$$

Figure 4 shows that the error remains relatively small but fluctuates over time due to discrete rebalancing.

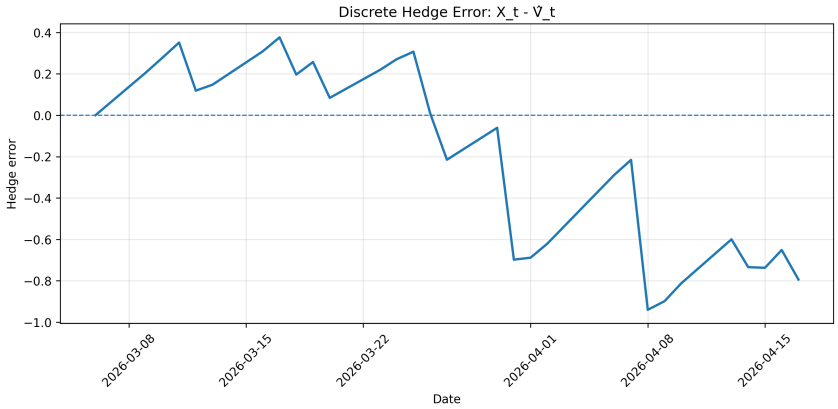


Figure 4: Hedge error $X_t - \hat{V}_t$ over the 30-day path.

5 Interpretation

The implied volatility is higher than the historical estimate, reflecting forward-looking market expectations of uncertainty. The delta evolves with both the underlying price and time to maturity, while the cash account adjusts to finance rebalancing.

The replicating portfolio tracks the theoretical Black–Scholes price closely but not perfectly. The discrepancy arises from discrete-time rebalancing: between trading dates, the hedge is not continuously updated, generating a tracking error.

Exercise 2. Stochastic Calculus and Pricing

2.1 From random walk to Brownian motion.

Let $\{\omega_i\}_{i\geq 1}$ be IID random variables such that

$$\mathbb{P}(\omega_i = 1) = \mathbb{P}(\omega_i = -1) = \frac{1}{2}, \qquad \mathbb{E}[\omega_i] = 0, \qquad \text{Var}(\omega_i) = 1.$$

Define the scaled random walk:

$$W^{(m)}(t) = \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor mt \rfloor} \omega_i.$$

Mean

Using linearity of expectation:

$$\mathbb{E}[W^{(m)}(t)] = \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor mt \rfloor} \mathbb{E}[\omega_i] = \frac{1}{\sqrt{m}} \sum_{i=1}^{\lfloor mt \rfloor} 0 = 0.$$

Variance

Since the ω_i are independent:

$$\mathrm{Var}\Big(W^{(m)}(t)\Big) = \frac{1}{m} \sum_{i=1}^{\lfloor mt \rfloor} \mathrm{Var}(\omega_i) = \frac{1}{m} \lfloor mt \rfloor.$$

As $m \to \infty$,

$$\frac{\lfloor mt \rfloor}{m} \to t.$$

Hence:

$$\mathrm{Var}(W^{(m)}(t)) \to t.$$

Increment from s to t

For $0 \leq s < t$:

$$W^{(m)}(t) - W^{(m)}(s) = \frac{1}{\sqrt{m}} \sum_{i=\lfloor ms \rfloor + 1}^{\lfloor mt \rfloor} \omega_i.$$

Therefore:

$$\mathbb{E}[W^{(m)}(t) - W^{(m)}(s)] = 0,$$

and

$$\mathrm{Var}(W^{(m)}(t) - W^{(m)}(s)) = \frac{\lfloor mt \rfloor - \lfloor ms \rfloor}{m} \to t - s.$$

Central Limit Theorem

The increment is the normalized sum of many IID centered variables. By the Central Limit Theorem:

$$W^{(m)}(t) - W^{(m)}(s) \xrightarrow[m \to \infty]{d} \mathcal{N}(0, t - s).$$

Motivation for Brownian Motion

This suggests defining a continuous-time limit process $\{W_t\}_{t \geq 0}$ such that:

- $W_0 = 0$,
- increments are Gaussian:
$$W_t - W_s \sim \mathcal{N}(0, t - s),$$
- increments over disjoint intervals are independent,
- sample paths are continuous.

This process is called **Brownian motion**.

2.2 Quadratic Variation

Let

$$0 = t_0 < t_1 < \cdots < t_n = t$$

be a partition of $[0, t]$, and define

$$\Delta t_i = t_{i+1} - t_i.$$

Set

$$Q_n(t) = \sum_{i=0}^{n-1} (W_{t_{i+1}} - W_{t_i})^2.$$

Expectation

Since Brownian increments satisfy

$$W_{t_{i+1}} - W_{t_i} \sim \mathcal{N}(0, \Delta t_i),$$

we have:

$$\mathbb{E}[(W_{t_{i+1}} - W_{t_i})^2] = \Delta t_i.$$

Hence:

$$\mathbb{E}[Q_n(t)] = \sum_{i=0}^{n-1} \Delta t_i = t.$$

Representation with Standard Normal Variables

Write

$$W_{t_{i+1}} - W_{t_i} = \sqrt{\Delta t_i}\, Z_i, \qquad Z_i \sim \mathcal{N}(0, 1).$$

Then:

$$(W_{t_{i+1}} - W_{t_i})^2 = \Delta t_i\, Z_i^2.$$

Since $Z_i^2 \sim \chi_1^2$, each squared increment is a time-step multiplied by a chi-squared random variable.

Variance of One Term

Using $\mathrm{Var}(Z_i^2) = 2$:

$$\mathrm{Var}\big((W_{t_{i+1}} - W_{t_i})^2\big) = (\Delta t_i)^2\, \mathrm{Var}(Z_i^2) = 2(\Delta t_i)^2.$$

Variance of $Q_n(t)$

Because Brownian increments are independent:

$$\mathrm{Var}(Q_n(t)) = \sum_{i=0}^{n-1} 2(\Delta t_i)^2.$$

Now:

$$\sum_{i=0}^{n-1} (\Delta t_i)^2 \le \left(\max_i \Delta t_i\right) \sum_{i=0}^{n-1} \Delta t_i = t \max_i \Delta t_i.$$

Thus:

$$\mathrm{Var}(Q_n(t)) \le 2t \max_i \Delta t_i \to 0 \quad \text{as} \quad \max_i \Delta t_i \to 0.$$

Since also $\mathbb{E}[Q_n(t)] = t$, we conclude:

$$Q_n(t) \to t$$

(in L^2, hence in probability).

Informal Differential Rule

This justifies the Itô rule:

$$(dW_t)^2 = dt.$$

Moreover:

$$dt^2 = o(dt), \qquad dt\, dW_t = o(dt),$$

because dW_t is of size $\sqrt{dt}$, so:

$$dt\, dW_t \sim dt^{3/2}, \qquad dt^2 \sim dt^2.$$

Both are negligible compared with dt, while $(dW_t)^2 \sim dt$ survives.

2.3 Black–Scholes with Inventory Costs

Assume:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

Let $V(S,t)$ be the derivative price.

Itô's Lemma

By Itô:

$$dV = V_t\, dt + V_S\, dS + \frac{1}{2} V_{SS} (dS)^2.$$

Since:

$$(dS)^2 = \sigma^2 S^2 dt,$$

we get:

$$dV = \left(V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right) dt + \sigma S V_S\, dW_t.$$

Replicating Portfolio

Let:

$$X_t = \Delta_t S_t + \Phi_t B_t, \qquad dB_t = r B_t dt.$$

Equivalently:

$$V_t - \Delta_t S_t = \Phi_t B_t.$$

These are the same statement: the option value equals holdings in stock plus holdings in bond.

Inventory Cost

Holding one unit of stock costs $cS_t dt$ per unit time, so:

$$dX_t = \Delta_t\, dS_t + \Phi_t\, dB_t - c\Delta_t S_t dt.$$

Substitute:

$$dX_t = \Delta_t(\mu S_t dt + \sigma S_t dW_t) + \Phi_t r B_t dt - c\Delta_t S_t dt.$$

Since:

$$\Phi_t B_t = V - \Delta_t S,$$

then:

$$dX_t = \Delta_t \mu S dt + \Delta_t \sigma S dW_t + r(V - \Delta_t S) dt - c\Delta_t S dt.$$

So:

$$dX_t = [rV + \Delta_t S(\mu - r - c)]\, dt + \Delta_t \sigma S dW_t.$$

Now impose replication: $dX_t = dV$.
Matching diffusion terms:

$$\Delta_t = V_S.$$

Matching drift terms:

$$V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} = rV + V_S S(\mu - r - c).$$

Simplify (the μ terms cancel):

$$V_t + (r + c)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

$$\boxed{V_t + (r + c)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0}$$

Comparison with Dividend Yield

Standard Black–Scholes with continuous dividend yield q:

$$V_t + (r - q)SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0.$$

Comparing coefficients:

$$r - q = r + c \quad \Longrightarrow \quad q = -c.$$

Thus inventory cost acts like a **negative dividend yield**.

2.4 Black–Scholes when $\ln S_t$ is Ornstein–Uhlenbeck

Assume:

$$d(\ln S_t) = \kappa(\theta - \ln S_t)dt + \sigma dW_t.$$

Let:

$$X_t = \ln S_t, \qquad S_t = e^{X_t}.$$

Dynamics of S_t

Apply Itô to $f(X) = e^X$:

$$f'(X) = e^X = S, \qquad f''(X) = e^X = S.$$

Hence:

$$dS_t = f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2.$$

Since $(dX_t)^2 = \sigma^2 dt$:

$$dS_t = S_t \left[\kappa(\theta - \ln S_t) + \frac{1}{2}\sigma^2\right] dt + \sigma S_t dW_t.$$

Pricing PDE

Write stock dynamics as:

$$dS_t = \mu(S_t, t)S_t dt + \sigma S_t dW_t,$$

where

$$\mu(S, t) = \kappa(\theta - \ln S) + \frac{1}{2}\sigma^2.$$

Using the same replication argument as before, the drift cancels out. Therefore the PDE is:

$$\boxed{V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV = 0}$$

Comparison with Standard Black–Scholes

This is the same PDE as the standard Black–Scholes equation (without dividends).

Although the *physical* stock drift is mean-reverting in log-price, arbitrage pricing depends on hedging away risk, not on the real-world expected return. Once the stochastic term is replicated using the stock, the drift disappears and only r remains.

Conclusion

The notation $o(dt)$ captures terms negligible relative to dt. In stochastic calculus, this matters because Brownian motion has nonzero quadratic variation:

$$(dW_t)^2 = dt.$$

This is the key difference from ordinary calculus. It produces the second-derivative term in Itô's lemma and therefore in pricing PDEs. Finally, even if the stock follows more complicated dynamics such as an Ornstein–Uhlenbeck log-price process, replication removes dependence on the physical drift, so the option pricing equation keeps the Black–Scholes structure.

Exercise 4. Option Trading Strategies

In this note we study six classical option strategies built from European calls, puts, stock, and cash. We assume the Black–Scholes framework and denote:

$$\omega = T - t$$

(time to maturity),

$$C(S, t; K, T)$$

the price of a European call with strike K and maturity T, and

$$P(S, t; K, T)$$

the price of a European put with strike K and maturity T.
For simplicity, we write:

$$C(K) := C(S, t; K, T), \qquad P(K) := P(S, t; K, T).$$

The key idea of the exercise is:

If a payoff is a sum of simpler payoffs, then its price is the same sum of prices.

This follows from no-arbitrage and linearity of replication.

4.1 Risk-Neutral Pricing

Under the risk-neutral probability measure $\mathbb{Q}$:

$$V(S, t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S].$$

Therefore:

$$V(S, t) = e^{-r\omega}\mathbb{E}^{\mathbb{Q}}[\Phi(S_T) \mid S_t = S].$$

Why this works for any European payoff

A European derivative pays only at time T, and its payoff depends only on S_T. Hence, once we know the distribution of S_T under $\mathbb{Q}$, we can price any payoff $\Phi(S_T)$ by taking its discounted expectation.

Distribution of S_T

Under Black–Scholes:

$$dS_u = rS_u\, du + \sigma S_u\, dW_u.$$

Applying Itô's lemma to $\ln S_u$:

$$d(\ln S_u) = \left(r - \frac{1}{2}\sigma^2\right) du + \sigma dW_u.$$

Integrating from t to T:

$$\ln S_T = \ln S + \left(r - \frac{1}{2}\sigma^2\right)\omega + \sigma(W_T - W_t).$$

Since:

$$W_T - W_t \sim N(0, \omega),$$

we obtain:

$$\ln S_T \sim N\left(\ln S + \left(r - \frac{1}{2}\sigma^2\right)\omega, \sigma^2\omega\right).$$

Thus S_T is lognormal.

Integral Form

If f_{LN} is the lognormal density:

$$V(S,t) = e^{-r\omega} \int_0^\infty \Phi(x)\, f_{LN}(x)\, dx.$$

4.2 Strategy Decomposition

We now analyze the six strategies listed in the assignment.

Important notation.

- Long call = buy one call.
- Short call = sell one call.
- Long put = buy one put.
- Long stock = buy one share.
- Short two calls = sell two call options.

Strategy	**Terminal Payoff** $\Phi(S_T)$	**How it is Built**	**Price at time** t
Straddle	$(S_T - K)^+ + (K - S_T)^+$	Long 1 call with strike K + Long 1 put with strike K	$C(K) + P(K)$
Covered Call	$S_T - (S_T - K)^+$	Long stock + Short 1 call with strike K	$S - C(K)$
Butterfly	$(S_T - K_1)^+ - 2(S_T - K_2)^+ + (S_T - K_3)^+$	Long call K_1, Short 2 calls K_2, Long call K_3	$C(K_1) - 2C(K_2) + C(K_3)$
Collar	$S_T + (K_P - S_T)^+ - (S_T - K_C)^+$	Long stock + Long put K_P + Short call K_C	$S + P(K_P) - C(K_C)$
Protective Put	$S_T + (K - S_T)^+$	Long stock + Long put	$S + P(K)$
Bull Call Spread	$(S_T - K_1)^+ - (S_T - K_2)^+$	Long call K_1 + Short call K_2	$C(K_1) - C(K_2)$

Table 3: Table 1: Summary of payoff, portfolio decomposition, and pricing formulas.

Explanation of the Main Constructions

Straddle

A straddle buys both a call and a put with the same strike.

$$\Phi = (S_T - K)^+ + (K - S_T)^+$$

If the stock goes far above K, the call pays. If the stock goes far below K, the put pays. Therefore the strategy benefits from *large moves in either direction.*

Price:

$$V = C(K) + P(K).$$

Covered Call

A covered call means owning the stock and selling a call.

$$\Phi = S_T - (S_T - K)^+$$

If $S_T \leq K$, the call expires worthless and you keep the stock. If $S_T > K$, the upside is partially given away because the sold call is exercised.

Price:

$$V = S - C(K).$$

Butterfly

The butterfly uses three strikes:

$$K_1 < K_2 < K_3$$

with K_2 usually centered.

Payoff:

$$\Phi = (S_T - K_1)^+ - 2(S_T - K_2)^+ + (S_T - K_3)^+.$$

It gains most when the final stock price stays near the middle strike K_2.

Price:

$$V = C(K_1) - 2C(K_2) + C(K_3).$$

Collar

A collar combines upside limitation and downside protection:

$$\Phi = S_T + (K_P - S_T)^+ - (S_T - K_C)^+.$$

Interpretation:

- The put creates a floor (protection below K_P).
- The short call caps gains above K_C.

Price:

$$V = S + P(K_P) - C(K_C).$$

Protective Put

A protective put means buying stock and buying insurance.

$$\Phi = S_T + (K - S_T)^+$$

If the stock crashes, the put offsets losses below K.
Price:

$$V = S + P(K).$$

Bull Call Spread

This strategy expresses a moderate bullish view.

$$\Phi = (S_T - K_1)^-(S_T - K_2)^+$$

(correct form:)

$$\Phi = (S_T - K_1)^+ - (S_T - K_2)^+, \qquad K_1 < K_2.$$

You gain when the stock rises, but profits are capped above K_2.
Price:

$$V = C(K_1) - C(K_2).$$

4.3 Closed-Form Black–Scholes Prices

We now compute each strategy explicitly.
The Black–Scholes formulas are:

$$C(K) = SN(d_1) - Ke^{-r\omega}N(d_2),$$

$$P(K) = Ke^{-r\omega}N(-d_2) - SN(-d_1),$$

where

$$d_1(K) = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)\omega}{\sigma\sqrt{\omega}}, \qquad d_2(K) = d_1(K) - \sigma\sqrt{\omega}.$$

When strikes differ, we compute d_1, d_2 using the corresponding strike.

Straddle

By definition:

$$V_{\text{str}} = C(K) + P(K).$$

Insert the formulas:

$$V_{\text{str}} = \big[SN(d_1) - Ke^{-r\omega}N(d_2)\big] + \big[Ke^{-r\omega}N(-d_2) - SN(-d_1)\big].$$

Group stock terms and bond terms:

$$V_{\text{str}} = S\big(N(d_1) - N(-d_1)\big) + Ke^{-r\omega}\big(N(-d_2) - N(d_2)\big).$$

Using:

$$N(-x) = 1 - N(x),$$

we obtain:

$$N(d_1) - N(-d_1) = 2N(d_1) - 1,$$

$$N(-d_2) - N(d_2) = 1 - 2N(d_2).$$

Therefore:

$$\boxed{V_{\text{str}} = S(2N(d_1) - 1) + Ke^{-r\omega}(1 - 2N(d_2))}$$

Covered Call

Price:

$$V_{\text{cc}} = S - C(K).$$

Substitute the call formula:

$$V_{\text{cc}} = S - \big[SN(d_1) - Ke^{-r\omega}N(d_2)\big].$$

Distribute the minus sign:

$$V_{\text{cc}} = S - SN(d_1) + Ke^{-r\omega}N(d_2).$$

Factor S:

$$\boxed{V_{\text{cc}} = S(1 - N(d_1)) + Ke^{-r\omega}N(d_2)}$$

Butterfly Spread

Price:

$$V_{\text{bf}} = C(K_1) - 2C(K_2) + C(K_3).$$

Now replace each call:

$$V_{\text{bf}} = \big[SN(d_1^{(1)}) - K_1e^{-r\omega}N(d_2^{(1)})\big]$$

$$-2\big[SN(d_1^{(2)}) - K_2e^{-r\omega}N(d_2^{(2)})\big]$$

$$+\big[SN(d_1^{(3)}) - K_3e^{-r\omega}N(d_2^{(3)})\big].$$

Collect stock terms:

$$S\Big(N(d_1^{(1)}) - 2N(d_1^{(2)}) + N(d_1^{(3)})\Big),$$

and discounted strike terms:

$$-e^{-r\omega}\Big(K_1N(d_2^{(1)}) - 2K_2N(d_2^{(2)}) + K_3N(d_2^{(3)})\Big).$$

Hence:

$$\boxed{V_{\text{bf}} = S\left[N(d_1^{(1)}) - 2N(d_1^{(2)}) + N(d_1^{(3)})\right] - e^{-r\omega}\left[K_1 N(d_2^{(1)}) - 2K_2 N(d_2^{(2)}) + K_3 N(d_2^{(3)})\right]}$$

where $d_i^{(j)} = d_i(K_j)$.

Collar

Price:

$$V_{\text{col}} = S + P(K_P) - C(K_C).$$

Substitute:

$$V_{\text{col}} = S + \left[K_P e^{-r\omega} N(-d_2^P) - SN(-d_1^P)\right] - \left[SN(d_1^C) - K_C e^{-r\omega} N(d_2^C)\right].$$

Expand:

$$V_{\text{col}} = S - SN(-d_1^P) - SN(d_1^C) + K_P e^{-r\omega} N(-d_2^P) + K_C e^{-r\omega} N(d_2^C).$$

Therefore:

$$\boxed{V_{\text{col}} = S\left[1 - N(-d_1^P) - N(d_1^C)\right] + e^{-r\omega}\left[K_P N(-d_2^P) + K_C N(d_2^C)\right]}$$

Protective Put

Price:

$$V_{\text{pp}} = S + P(K).$$

Insert put formula:

$$V_{\text{pp}} = S + Ke^{-r\omega} N(-d_2) - SN(-d_1).$$

Factor S:

$$\boxed{V_{\text{pp}} = S\big(1 - N(-d_1)\big) + Ke^{-r\omega} N(-d_2)}$$

Since $1 - N(-d_1) = N(d_1)$, equivalently:

$$\boxed{V_{\text{pp}} = SN(d_1) + Ke^{-r\omega} N(-d_2)}$$

Bull Call Spread

Price:

$$V_{\text{bull}} = C(K_1) - C(K_2).$$

Substitute:

$$V_{\text{bull}} = \left[SN(d_1^{(1)}) - K_1 e^{-r\omega} N(d_2^{(1)})\right] - \left[SN(d_1^{(2)}) - K_2 e^{-r\omega} N(d_2^{(2)})\right].$$

Expand:

$$V_{\text{bull}} = S\big(N(d_1^{(1)}) - N(d_1^{(2)})\big) - e^{-r\omega}\big(K_1 N(d_2^{(1)}) - K_2 N(d_2^{(2)})\big).$$

Thus:

$$\boxed{V_{\text{bull}} = S\left[N(d_1^{(1)}) - N(d_1^{(2)})\right] - e^{-r\omega}\left[K_1 N(d_2^{(1)}) - K_2 N(d_2^{(2)})\right]}$$

4.4 Greeks

Because derivatives are linear operators:

$$G(aV_1 + bV_2) = aG(V_1) + bG(V_2),$$

for any Greek G.
So the Greeks of each strategy are obtained by summing the Greeks of its components.

Strategy	**Greeks**
Straddle	$\Delta = \Delta_C + \Delta_P, \quad \Gamma = \Gamma_C + \Gamma_P, \quad \Theta = \Theta_C + \Theta_P, \quad \nu = \nu_C + \nu_P, \quad \rho = \rho_C + \rho_P$
Covered Call	$\Delta = 1 - \Delta_C, \quad \Gamma = -\Gamma_C, \quad \Theta = -\Theta_C, \quad \nu = -\nu_C, \quad \rho = -\rho_C$
Butterfly	$\Delta = \Delta_1 - 2\Delta_2 + \Delta_3$ $\Gamma = \Gamma_1 - 2\Gamma_2 + \Gamma_3$ $\Theta = \Theta_1 - 2\Theta_2 + \Theta_3$ $\nu = \nu_1 - 2\nu_2 + \nu_3$ $\rho = \rho_1 - 2\rho_2 + \rho_3$
Collar	$\Delta = 1 + \Delta_P - \Delta_C$ $\Gamma = \Gamma_P - \Gamma_C$ $\Theta = \Theta_P - \Theta_C$ $\nu = \nu_P - \nu_C$ $\rho = \rho_P - \rho_C$
Protective Put	$\Delta = 1 + \Delta_P, \quad \Gamma = \Gamma_P, \quad \Theta = \Theta_P, \quad \nu = \nu_P, \quad \rho = \rho_P$
Bull Call Spread	$\Delta = \Delta_1 - \Delta_2$ $\Gamma = \Gamma_1 - \Gamma_2$ $\Theta = \Theta_1 - \Theta_2$ $\nu = \nu_1 - \nu_2$ $\rho = \rho_1 - \rho_2$

Table 4: Greeks obtained by linearity.

4.5 Interpretation

Strategy	Market View	Gamma Profile	Typical Use Case
Straddle	Volatility-based	Long gamma	Expecting a strong move but unsure of direction
Covered Call	Mild bullish / neutral	Short gamma	Generate extra income on an existing stock position
Butterfly	Range-bound	Mixed	Expecting price near a target level
Collar	Protective bullish	Mixed / limited	Keep stock exposure while reducing downside risk
Protective Put	Protective bullish	Long gamma	Insurance for a long-term stock investor
Bull Call Spread	Moderately bullish	Positive but capped	Cheaper bullish position than buying a naked call

Table 5: Interpretation of the six strategies.

Conclusion

The whole exercise is based on one fundamental principle:

$$\boxed{\text{Complex payoffs can be decomposed into simple payoffs.}}$$

Once we know how to price calls, puts, stock, and bonds under Black–Scholes, every strategy follows immediately by linearity. The same reasoning applies to sensitivities (Greeks): portfolio risk is simply the sum of the risks of each component.